

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech.

Branch : AI&DS & Allied

Semester : III

Subject Code & Name: BTAIC302 An Introduction to Artificial Intelligence

Max Marks: 60

Date: 02/07/2024

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Answer Any Two of the following questions.		
A) Define Artificial Intelligence and Explain The Need of AI	L1/CO1	6
B) Explain Turing Test and it's significance	L2/CO1	6
C) What is Agent? Explain different Types of Agent	L3/CO1	6
Q.2 Answer Any Two of the following questions.		
A) Explain the terms 1) State Space 2) Generate and Test with ref. to AI	L1/CO2	6
B) Draw suitable diagram and describe following issues of Hill Climbing Algorithm:- a) Local Maxima b) Plateaus	L2/CO2	6
C) Describe Breadth First Search Algorithm, demonstrate its source code in Python	L3/CO2	6
Q.3 Answer Any Two of the following questions.		
A) Explain Dijkstra Algorithm in detail	L1/CO3	6
B) Illustrate the concept of Ant Colony Optimization with neat explanation	L2/CO3	6
C) What is Neural Network, Explain in Detail	L3/CO3	6
Q.4 Answer Any Two of the following questions.		
A) What is Minimax Algorithm? Explain with Example	L1/CO4	6
B) Explain SSS* Algorithm	L2/CO4	6
C) Explain the concept of Branch and Bound Algorithm	L3/CO4	6
Q.5 Answer Any Two of the following questions.		
A) Explain Constraint Satisfaction Problem with it's various components.	L1/CO5	6
B) Define i) Lookahead strategies ii) Strategic retreat	L2/CO5	6
C) Compare backtracking Algorithm and Branch and Bound Algorithm	L3/CO5	6

***** End *****